Concept-to-Animation: LLM-Generated Manim Explainer Videos

CSE 434 Midterm Check-in Report

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Team: Concept-to-Animation **Roles:** *Jonah* — baseline pipeline (LLM client, planner, code generation, render harness, logging, ablation runner), repair and layout loops, analysis, demo. *Koray* — few-shot curation, RAG corpus and retriever, narration. *Both* — human evaluation and reports.

I. Project Overview

Given a short description of a math or CS concept (e.g. "binary search"), our system writes a short explainer video in the style of 3Blue1Brown by having an LLM plan a storyboard and then write Manim Community Edition (ManimCE) code for each scene. LLM-written animation code fails in two characteristic ways: it calls *hallucinated or outdated Manim APIs* (the code crashes), and it produces *bad layouts* (objects overlap or leave the frame). Our research question is: **which generative techniques make LLM-written animation code render reliably and look clear?** We answer it with a cumulative ablation that starts from a zero-shot baseline and adds one technique at a time: error-driven self-repair, few-shot examples, retrieval-augmented generation (RAG) over the ManimCE docs, and a layout-repair loop driven by a bounding-box checker. This connects directly to course topics on prompting, in-context learning, RAG, iterative self-refinement, and LLM code generation.

II. Dataset

Concepts. We curated 6 concepts in three difficulty tiers of two each, chosen by how directly the idea maps to pictures: *concrete* (Pythagorean theorem, derivative as slope), *structural* (matrix multiplication, binary search), and *abstract* (recursion, Big-O notation). Each has a one-sentence prompt; the tiers let us ask whether performance drops as concepts get less visual.

Storyboards. For each concept the planner produces a storyboard of 3–4 scenes (20 scenes in total), each with a title, a visual description and narration. Storyboards are validated against a schema and cached, so every condition generates code for exactly the same scenes and only the code-generation step differs. All 6 were produced in one batch on 2026-11-10 by the same model.

Retrieval corpus (planned): the ManimCE docs and example gallery (MIT license), pinned to our render version (v0.22.1), chunked and embedded for RAG.

Ethics. The data contains no personal information. The main risk is that a fluent, polished video can teach *wrong* math; render success does not catch this, so we add human ratings of mathematical correctness. Generated videos will be labeled as AI-generated.

III. Methods

Figure 1 shows the current system (v1). A **planner** turns the concept into a storyboard. A **code generator** writes one ManimCE `Scene` subclass per storyboard scene; its prompt states the pinned Manim version, forbids the older ManimGL library, and gives the frame bounds ($x \in [-7.1, 7.1]$, $y \in [-4, 4]$). Post-processing is deliberately minimal (extract the code block, rename the class) so it cannot silently fix the errors we are measuring. Each scene is **rendered** in a separate subprocess with a 120 s timeout; we check that the expected video file was actually written, because Manim exits successfully even when the named scene does not exist. Failures are **classified** from the traceback into `hallucinated_api`, `bad_signature`, `latex`, `timeout`, `syntax` or `other`.

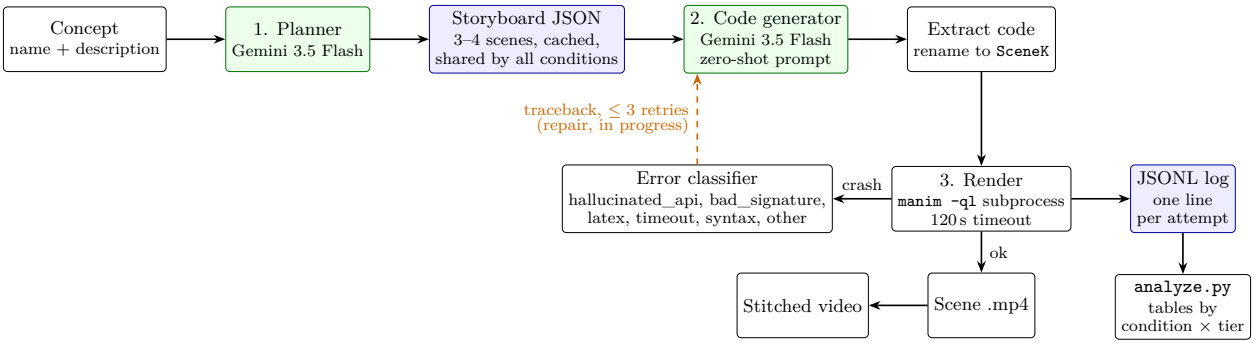


Figure 1: System design v1. Green: LLM calls; blue: stored artifacts. Solid path: the zero-shot baseline. Dashed: the error-repair loop (`zero_shot_repair`), which feeds the traceback back to the code generator. Few-shot, RAG and layout repair will attach to the code generator later.

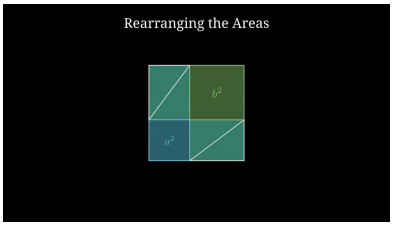
One subtlety: Manim's `Mobject` invents any `get_*` method on demand, so a hallucinated call like `axes.get_secant_line(...)` fails with a `TypeError` rather than an `AttributeError`; our classifier recognizes this pattern as a hallucinated API. Every attempt is logged as one JSONL line (tokens, latency, error, code path), and responses are cached so reruns cost nothing.

Experimental design. The full study is a cumulative ablation over six conditions: `zero_shot`, `zero_shot_repair`, `few_shot`, `rag`, `rag_repair` and `rag_repair_layout`. Temperature is fixed at 0.2, conditions without repair get exactly one render attempt, and repair conditions get up to three extra attempts. Metrics are computed per scene and aggregated by condition and tier: render success, first-try success, average repair iterations, layout-defect rate, error types, and two human scores (correctness and clarity, 1–5, blind to condition; a failed scene scores 0, so human scores reflect the whole pipeline).

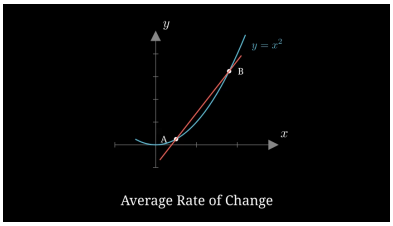
Changes since the proposal (approved by Dr. Zhang). (1) *Gemini 3.5 Flash instead of GPT + an open model:* we tried Qwen2.5-Coder (7B and 3B, 4-bit via Ollama), but its Manim code was too poor to make the comparison informative; open models stay available behind our LLM interface. (2) *No LangChain:* plain Python lets us log every prompt, reply and token count exactly. (3) *6 concepts instead of 40,* so every condition can be run and every video hand-rated. Google bills `gemini-3.5-flash` against a `gemini-3.6-flash` quota, so the name may be an alias; we log the provider-reported model version on every call to detect drift.

IV. Preliminary Results

The baseline pipeline runs end to end. A first zero-shot run stopped at the Gemini free tier's 20-requests-per-day limit after 11 of 20 scenes (8 rendered); full runs use a paid key. Table 1 reports both conditions on all 20 scenes.



(a) Pythagorean theorem, scene 3 (rendered).



(b) Derivative as slope, scene 1 (rendered).

Figure 2: Final frames from two zero-shot scenes that rendered on the first try.

	Automatic			Human (0–5; failed scene = 0)	
Condition	Render success	First-try success	Avg. repairs	Correctness	Clarity
`zero_shot`	? (?/20)	?	–	?	?
concrete (7)	?	?	–	?	?
structural (6)	?	?	–	?	?
abstract (7)	?	?	–	?	?
`zero_shot_repair`	? (?/20)	?	?	?	?
concrete (7)	?	?	?	?	?
structural (6)	?	?	?	?	?
abstract (7)	?	?	?	?	?

Table 1: Gemini 3.5 Flash, all 20 scenes (scenes per tier in parentheses). For `zero_shot_repair`, render minus first-try success is what repair adds; avg. repairs is over scenes that eventually rendered. Human scores: mean of both raters over all 20 scenes; rendered scenes rated 1–5 blind to condition, failed scenes scored 0.

Qualitative examples. *Success:* all four Pythagorean-theorem scenes rendered on the first try. Scene 3 (Fig. 2a) rearranges four copies of the triangle inside an $(a + b)$ square to expose the a^2 and b^2 regions, a clean and correct version of the classic proof. *Contrasting failure:* for the derivative, scene 1 (Fig. 2b) correctly draws a secant through two points on $y = x^2$, but scenes 2 and 3, which must *animate* the secant approaching the tangent, both crashed. The model reached for convenience methods (`get_secant_line`, `get_tangent_line`) that sound right but are not in ManimCE. Static drawings succeed; scenes that need an updater-driven animation are where hallucinated APIs appear.

Error types. Every failure so far is API misuse, never syntax or LaTeX: invented methods (`hallucinated_api`) or keyword arguments the real class lacks (`bad_signature`, e.g. `grid_stroke_width`). Each error message names the bad call, so these are exactly what traceback-driven repair should fix and RAG over the real docs should prevent.

Human ratings. [TODO: mean correctness and clarity per condition (failed scenes = 0), inter-rater agreement, and what the ratings show that render success does not.]

V. Next Steps

- **Few-shot and RAG** (Koray): 3–5 curated modern ManimCE example scenes, and a retriever over the pinned ManimCE docs checked on whether it returns the right classes (e.g. `Axes.plot` for graphs).
- **Layout checker and repair:** snapshot bounding boxes after each animation, flag off-frame objects and unintended overlaps, and describe defects to the LLM in plain language; tune the threshold on about ten hand-labeled scenes.
- **Evaluation:** the full six-condition ablation by tier, token cost and latency per condition, human ratings with inter-rater agreement, and a success/failure gallery. **Demo:** gTTS narration and a type-a-concept web demo.

Open questions for feedback. Are 20 scenes enough to separate conditions? Is a bounding-box check a fair proxy for clarity, or should clarity rest on human ratings alone? Should we add leave-one-out rows to check the ablation's stacking order?